

ANSIBLE COMMANDS CHEATSHEET

(A COMPLETE QUICK-REFERENCE GUIDE)



1. Notation & Conventions

- `<angle-brackets>` : placeholders (e.g., `<inventory>`, `<playbook>`)
- `[square-brackets]` : optional flags
- `-i <inventory>` : specify inventory file
- `-m <module>` : specify module
- `-a "<args>"` : module arguments
- `-b` : become (sudo)
- `--check` : dry-run (no changes applied)
- `-v / -vv / -vvv` : verbosity levels

2. Basics & Setup

- `ansible --version` – Show Ansible version.
- `ansible-config dump` – Show current configuration.
- `ansible-config view` – View `ansible.cfg`.
- `ansible-doc -l` – List all modules.
- `ansible-doc <module>` – Module documentation.
- `ansible-inventory --list` – Show inventory as JSON.
- `ansible-inventory --graph` – Inventory graph view.

3. Inventory Manage hosts and groups.

- `ansible all -i hosts --list-hosts` – List all hosts.
- `ansible <group> -i hosts --list-hosts` – List group hosts.
- `ansible all -m ping` – Test connectivity.
- `ansible all -m ping -u <user>` – Ping using specific user.

4. Ad-Hoc Commands

Run one-time commands without playbooks

- `ansible all -m ping` – Ping all hosts.
- `ansible all -m shell -a "uptime"` – Run shell command.
- `ansible all -m command -a "df -h"` – Run command module.
- `ansible all -m copy -a "src=a.txt dest=/tmp/a.txt"` – Copy file.
- `ansible all -m file -a "path=/tmp/test state=touch"` – Create file.
- `ansible all -m yum -a "name=nginx state=present" -b` – Install package.
- `ansible all -m service -a "name=nginx state=started" -b` – Start service.

5. Playbooks

Execute automation workflows

- `ansible-playbook playbook.yml` – Run playbook.
- `ansible-playbook playbook.yml -i hosts` – Run with inventory.
- `ansible-playbook playbook.yml --check` – Dry-run.
- `ansible-playbook playbook.yml --syntax-check` – Syntax check.
- `ansible-playbook playbook.yml --list-tasks` – List tasks.
- `ansible-playbook playbook.yml --list-hosts` – List target hosts.
- `ansible-playbook playbook.yml -v` – Verbose output.

6. Modules – Common

Package Management

- `yum`, `dnf`, `apt` – Manage packages.

Files

- `copy` – Copy files.
- `template` – Jinja2 templates.

- [file](#) – File & directory management.
- [fetch](#) – Fetch files from remote hosts.

System

- [service](#) – Manage services.
- [user](#) – Manage users.
- [group](#) – Manage groups.
- [cron](#) – Manage cron jobs.

7. Variables & Facts

Manage dynamic values & Gather system information

- [ansible all -m setup](#) – Gather facts.
- [ansible all -m setup -a "filter=ansible_os_family"](#) – Filter facts.
- [ansible-playbook play.yml -e "var=value"](#) – Extra variables.
- [ansible-playbook play.yml -e @vars.yml](#) – Load vars from file.

8. Roles

Reusable automation components

- [ansible-galaxy init <role>](#) – Create role structure.
- [ansible-galaxy role list](#) – List installed roles.
- [ansible-galaxy role install <role>](#) – Install role.
- [ansible-galaxy role remove <role>](#) – Remove role.
- [ansible-galaxy search <keyword>](#) – Search roles.

9. Ansible Galaxy

Community roles & collections

- [ansible-galaxy collection list](#) – List collections.

- `ansible-galaxy collection install <name>` – Install collection.
- `ansible-galaxy collection remove <name>` – Remove collection.
- `ansible-galaxy collection search <keyword>` – Search collections.

10. Vault (Secure Secrets Management)

- `ansible-vault create secrets.yml` – Create encrypted file.
- `ansible-vault edit secrets.yml` – Edit encrypted file.
- `ansible-vault view secrets.yml` – View encrypted file.
- `ansible-vault encrypt file.yml` – Encrypt file.
- `ansible-vault decrypt file.yml` – Decrypt file.
- `ansible-playbook play.yml --ask-vault-pass` – Run with vault password.

11. Handlers & Tags

Control task execution

- `ansible-playbook play.yml --tags install` – Run specific tags.
- `ansible-playbook play.yml --skip-tags config` – Skip tags.
- `notify:` – Trigger handlers.
- `handlers:` – Define handlers.

12. Debugging & Troubleshooting

Troubleshoot execution issues

- `ansible-playbook play.yml -vvv` – Detailed debug.
- `ansible all -m debug -a "var=ansible_hostname"` – Debug variable.
- `ansible-playbook play.yml --start-at-task "<task-name>"` – Resume from task.
- `ansible-playbook play.yml --step` – Step-by-step execution.

13. Privilege Escalation

Run tasks with elevated permissions

- `-b` – Enable become (sudo).
- `--become-user <user>` – Run as specific user.
- `--ask-become-pass` – Ask sudo password.
- `become: true` – Enable in playbook.

14. Loops & Conditionals

- `loop: / with_items:` – Loop tasks.
- `when:` – Conditional execution.
- `register:` – Store output in variable.
- `changed_when:` – Override changed status.
- `failed_when:` – Override failure conditions.

15. Cleanup (Manual)

Local cleanup operations.

- `rm -rf ~/.ansible/tmp` – Clear Ansible temp files
- `rm -rf roles/*` – Remove downloaded roles

16. Ansible Workflow (One-Line Summary)

`inventory → ping → ad-hoc → playbook → roles → vault → deploy`